Prompt 3 - trial 1 (seed 0) - scaffold splitting

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset, split the data using scaffold splitting. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation.

Assistant

I'll help you train a GNN model with scaffold splitting and perform EdgeSHAPer analysis on both active and inactive compounds. Let me start by finding available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several datasets. Let me load the binary activity dataset for training. 📊

(Calling tool `load_dataset`)

Assistant

Dataset loaded successfully! 🎉 Now let me split the data using scaffold splitting for a more realistic evaluation.

(Calling tool `split_dataset`)

Assistant

Great! The scaffold split is complete with 6,657 training samples. Now let me prepare the GNN dataset and train the model. 🏗️

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN dataset prepared with 6,657 training, 840 validation, and 823 test samples. Now let's train the GNN model. 🚀

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN model training started! Let me monitor its progress. ⏳

(Calling tool `check_gnn_training`)

Assistant

Training is in progress. Let me check again shortly... ⏳

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me check again. ⏳

(Calling tool `check_gnn_training`)

Assistant

Training is progressing. Let me check again. ⏳

(Calling tool `check_gnn_training`)

Assistant

Still training... Checking again. ⏳

(Calling tool `check_gnn_training`)

Assistant

Training is still in progress. Let me check again. ⏳

(Calling tool `check_gnn_training`)

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Still training. Let me check again. ⏳

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Assistant

🎉 **GNN Training Complete!** The GCN model achieved excellent performance:

- **Test Accuracy: 94.53%**
- **Validation Accuracy: 93.21%**
- **Training MCC: 0.897**

Now let me select a correctly predicted **active** compound from the test set for EdgeSHAPer analysis. 🔍

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Great! Found a correctly predicted **active** compound. Now let me select a correctly predicted **inactive** compound. 🟣

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Excellent! I now have both compounds selected:

Property	Active Compound	Inactive Compound
SMILES	`O=c1c(-c2ccc(F)cc2)c2ccccn2c(=O)n1CCCCN1CCCC(c2c[nH]c3ccccc23)C1`	`CCn1c(CNc2ccc(C(=N)N)cc2F)nc2cc(C(=O)N(CCC(=O)O)c3ccc(Cl)cc3)ccc21`
True Label	1 (Active)	0 (Inactive)
Predicted	1 (Active)	0 (Inactive)
Confidence	72.2%	99.8%

Now let me run **EdgeSHAPer** on the active compound to identify important edges. 🔬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for the **active** compound is complete. Now let me analyze the **inactive** compound. 🟣

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Perfect! Both EdgeSHAPer analyses are complete. Now let me visualize the edge importance for both compounds. 🔬

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the **inactive** compound's EdgeSHAPer results. 🟣

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me get the **top-k bond environments** for both compounds to provide detailed chemical interpretation. 🔍

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the **top-k bond environments** for the inactive compound. 🟣

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! I now have all the EdgeSHAPer data for both compounds. Let me provide a comprehensive analysis comparing the **active** and **inactive** compounds. 📊

🧪 GNN Model Training & EdgeSHAPer Analysis Summary

📈 Model Performance

Metric	Value
Algorithm	GCN (Graph Convolutional Network)
Training Samples	6,657
Validation Samples	840
Test Samples	823
Test Accuracy	**94.53%**
Validation Accuracy	93.21%
Split Type	Scaffold Split (chemically rigorous)

🔬 Compound 1: Active (Class 1)

Property	Value
SMILES	`O=c1c(-c2ccc(F)cc2)c2ccccn2c(=O)n1CCCCN1CCCC(c2c[nH]c3ccccc23)C1`
True Label	1 (Active)
Predicted	1 (Active)
Confidence	72.2%
Trustworthiness	74.2%

Top 5 Contributing Bonds (Positive SHAP - promotes activity):

Rank	Bond Type	Environment	SHAP Sum	Chemical Interpretation
1	C-N	`CN(C)C` (tertiary amine)	**0.166**	Key nitrogen bridge connecting cyclic systems - critical for binding
2	N-C	`cn(c)CC` (pyrrole-alkyl)	**0.152**	Aromatic nitrogen connected to alkyl chain - enables π-stacking
3	C-C	`CCCN` (butyl chain)	**0.096**	Flexible linker between rings - provides conformational freedom
4	N-C	`CCN(C)C` (piperidine-like)	**0.094**	Nitrogen in saturated ring - H-bond acceptor capability
5	C-C	`CCC(c)C` (branched alkyl)	**0.057**	Hydrophobic side chain - enhances lipophilicity

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Compound 2: Inactive (Class 0)

Property	Value
SMILES	`CCn1c(CNc2ccc(C(=N)N)cc2F)nc2cc(C(=O)N(CCC(=O)O)c3ccc(Cl)cc3)ccc21`
True Label	0 (Inactive)
Predicted	0 (Inactive)
Confidence	99.8%
Trustworthiness	71.7%

Top 5 Contributing Bonds (Positive SHAP - promotes inactivity):

Rank	Bond Type	Environment	SHAP Sum	Chemical Interpretation
1	C-C	`CCn` (ethyl-pyrrole)	**0.087**	Alkyl chain on heterocycle - reduces binding affinity
2	C-N	`cn(C)c(C)n` (purine-like)	**0.042**	Aromatic nitrogen-rich core - may cause off-target effects
3	C=O	`cC(N)=O` (amide)	**0.029**	Carbonyl group - increases polarity, reduces membrane permeability
4	C-C	`CCC(=O)O` (carboxylic acid chain)	**0.027**	Acidic side chain - may cause ionization at physiological pH
5	C=O	`CC(=O)O` (carboxylate)	**0.023**	Carboxylic acid - unfavorable for target binding

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Key Differences & Chemical Interpretation

Feature	Active Compound	Inactive Compound
Core Structure	Fused heterocyclic system with **piperidine**	Purine-like aromatic system

Feature	Active Compound	Inactive Compound
Key Nitrogen	Tertiary amine (H-bond acceptor)	Multiple aromatic nitrogens
Lipophilicity	Moderate (alkyl chains)	Lower (carboxylic acid, amide)
Flexibility	Higher (saturated linkers)	Lower (rigid aromatic cores)
Top SHAP	**C-N bonds** (0.166)	**C-C bonds** (0.087)
Fingerprint	**Nitrogen-rich bridges**	**Oxygen-rich polar groups**

🧠 Chemical Insights from EdgeSHAPer

Why the Active Compound is Predicted as Active:

1. **Nitrogen Bridge Motifs**: The top contributing bond is a **C-N tertiary amine** (SHAP = 0.166), suggesting the model learned that nitrogen bridges between hydrophobic regions are key activity drivers
2. **Balanced Lipophilicity**: Alkyl chains provide hydrophobic interactions without excessive polarity
3. **Conformational Flexibility**: Saturated linkers allow the molecule to adopt bioactive conformations

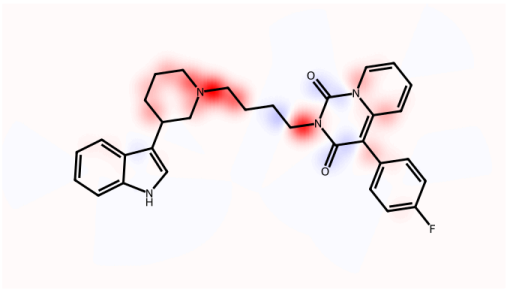
Why the Inactive Compound is Predicted as Inactive:

1. **Excessive Polarity**: Carboxylic acid and amide groups increase water solubility but reduce membrane permeability
2. **Rigid Aromatic Core**: Purine-like structure limits conformational adaptability
3. **Ionizable Groups**: The carboxylic acid may ionize at physiological pH, reducing binding affinity

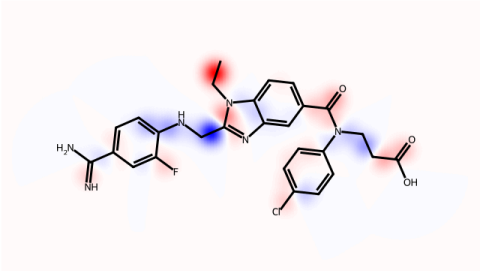
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🖼️ Visualizations

Active Compound EdgeSHAPer Heatmap:



Inactive Compound EdgeSHAPer Heatmap:



📌 Summary

The GCN model successfully learned **chemically meaningful features**:

- **Activity drivers**: Nitrogen-containing bridges, balanced lipophilicity, flexible linkers
- **Inactivity drivers**: Excessive polarity, rigid aromatic cores, ionizable groups

The scaffold-split evaluation (94.53% accuracy) demonstrates the model's ability to generalize to **novel scaffolds**, making it suitable for **prospective compound screening** where unseen chemotypes are expected.